

TU Berlin Chair Agro Mechatronics Prof. Dr.-Ing. C. Weltzien				Requirements Compliance Table for Modul Mobile Working Robotic Systems		Team Members: I. Zytoon, N. Aschenbroich, J. Korn	
				Requirements	Score	Fulfillment	
No.	Date:	Origin	D/R (F/W)	Is rated on a system from A to D			
G01	28.04.25	C	D	General Requirements Robust and safe design	B	Robust design with clear hierarchy and distributed architecture, basic safety features	
G02	28.04.25	C	D	Drivable omnidirectional robot platform	D	Currently undrivable, as robot is still unassembled	
G03	28.04.25	C	D	Drive-by-Wire control	B	Communication and control has been implemented but not yet tested on the final robot	
G04	28.04.25	C	D	Interface definition to/from navigation control	A	See documentary	
G05	28.04.25	C	D	Fitting, robust & cost-efficient selection of electrical components	A	See documentary and parts list	
G06	28.04.25	C	D	Design for agricultural application, in remote locations	B	Bluetooth/WIFI connection required to control robot, hardware is not fully waterproof	
G07	9.5.2025	IZ	D	Use existing platform and motors (NEMA23 stepper, 57BLF03 BLDC)	A	Modularity and extendibility were achieved for software and hardware	
G08	9.5.2025	IZ	D	Compact, modular design for maintainability and debug access	A		
G09	28.04.25	C	D	Completion of project until 16.07.25 , submission of documentation until 13.08.25	A		
				Specific Requirements (In Progress)			
S01	28.04.25	C	R	Remote control with MIT Inventor App via Bluetooth/WIFI (Raspberry Pi needed)	A	Successful connection with extensive testing	
S02	28.04.25	JK	D	Multiple drive modes with flexible remote mode switching	A	3 distinct drive modes, selectable in user interface, flexible switching	
S03	28.04.25	JK	R	Fail-safe for loss of connection (e.g. ROS ping messages)	B	Fail safe on Raspberry Pi stops the robot, but no error handling	
S04	28.04.25	C	R	Integration of hardware-level safety functionality (Emergency Stop)	D	No E-Stop yet, tbd!	
S05	28.04.25	JK	R	ROS2 interface for future integration of advanced autonomy (e.g. path planning, mapping)	A	ROS interfaces designed and documented, easily extendible	
				Component Enclosure Requirements			
C01	09.05.25	JK	D	Robust enclosure for Raspberry Pi, STM32 board, converters, and interfaces	A	Enclosure designed, printed, and assembled, structural integrity assessed but not to failure	
C02	09.05.25	JK, IZ	R	3D printed from ABS, modular layout for additional interface boards and sensors	A	Printed from PETG, better suitable than ABS, modular layout	
C03	09.05.25	JK	R	Screw cap, easy access for debugging and maintenance	A	Screw cap, removable driver plates	
C04	09.05.25	JK	R	Cable entries with sealed grommets	D	No sealed cable entries yet	
C05	09.05.25	JK	D	Heat management	B	Passive heat management, no aluminium in the prototype	

Process Requirements					
P01	28.04.25	C	D	Meet all deadlines and due dates	A
P02	28.04.25	C	D	Keep to the house rules	A
P03	28.04.25	C	D	Respectful and careful handling of facilities, infrastructure, tools in student working room	A
P04	28.04.25	JK	D	Extensive and timely documentation of work and interface definition	A See documentation

A – Full fulfillment; B – Mostly satisfactory fulfillment, C – Insufficient fulfillment, D – No fulfillment